

DATA STRUCTURES & ALGORITHMS (DSA USING C)



C Language Contents:

- Introduction to C Language
- Role of Compilers and Assemblers
- Procedural programming Approach
- Top to Bottom Approach

Introduction to C Basic

- Keywords
- Data Types
- Variables
- Constants
- Iden
- fiers
- Tokens
- Operators
- Memory Management in program

Flow Control Statements

- Jump Statements
- Goto
- Break
- Continue
-
- Conditional Statements
- If statement
- If else statement
- Nested if else
- If else if ladder
- Switch Case statement
- Interview Questions
-
- Iteration statements
- For loop
- While Loop
- Do while loop
- Nested loops
- Pattern Programming
- Interview Questions

Functions

- Introduction to Functions
- Types of Functions

- Call by Value and Reference
- Recursion
- Various Examples of Recursion
- Storage classes
- Interview Questions

Arrays

- Introduction to Arrays
- Several examples of Arrays like insert, delete, counter occurrence of items etc.
- Searching
- Sorting
- 2 D Array
- Several examples of 2D arrays
- Multidimensional Arrays

Strings

- Introduction to Strings
- String Handling Built in functions
- Several examples of strings
- Array of strings

Pointers

- Introduction to Pointers
- Pointer Expressions
- Types of Pointers
- Various Examples of pointers
- Array using pointers
- String using pointers
- Pointers in Functions

Dynamic Memory Allocation

- Introduction to Dynamic Memory Allocation
- Advantages of Dynamic Memory Allocation
- malloc(), calloc(), realloc(), free()
- Array Implementation using Dynamic Memory allocation

Structure

- Introduction to structure
- Advantages of Structure
- Array of Structure
- Structure using Pointer
- Structure with Functions
- Applications of Structures

Union

- Introduction to union
- Advantages of union
- Difference between union and structure

File Handling

- Introduction to file management
- File opening modes

- Opening and closing a file
- Input Output operations on file
- Applications of File management Project

Macro and Compiler Control Directives

- #define
- #undef
- #ifdef
- #elif

DATA STRUCTURE AND ALGORITHMS CONTENT (using C)

- **Introduction to Data Structure and Algorithms**
- Analysis of Algorithm
- Order of Growth
- Asymptotic Notations
- Big O Notation
- Omega Notation
- Theta Notation
- Analysis of common loops
- Space Complexity
- Practice Problems

Mathematics

- Mathematics
- Count Digits
- Palindrome Numbers
- Factorial of Numbers
- GCD of Two Numbers
- LCM of Two Numbers
- Check For Prime
- Prime Factors
- Computing Power
- Practice Problems

Recursion

- Introduction to Recursion
- Applications of Recursion
- Various Problems on Recursion
- Checking Palindrome
- Sum of Digits
- Tower of Hanoi Problem
- Practice Problems

Arrays

- Introduction and Advantages
- Types of Arrays
- Operations on Arrays
- Searching
 - - Binary Search
 - - Linear Search
- Sorting

- Bubble Sort
- - Insertion Sort
- Selection Sort
- Merge Sort
- Quick Sort
- Heap Sort
- Counting Sort
- - Radix Sort
- - Bucket Sort
- Insertions
- Deletion
- Reversing
- Practice Problems

Hashing

- Application of Hashing
- Working and examples on various Hash Functions
- Introduction and various techniques on Collision Handling
- Chaining and its implementation
- Chaining v/s Open Addressing
- Double Hashing
- Practice Problems

Strings

- Discussion of String DS
- Strings in C
- Rabin Karp Algorithm
- KMP Algorithm
- Practice Problems

Linked Lists

- Introduction
- Doubly Linked List
- Circular Linked List
- Practice Problems

Stacks

- Understanding the Stack Data Structure
- Applications of Stack
- Implementation of Stack in Array and Linked List
- Infix, Prefix and Postfix Introduction
- Practice Problems

Queue

- Introduction and Application
- Implementation of Queue in Array and Linked List
- Priority Queue
- Practice Problems

Tree

- Introduction and Application
- Tree Traversal

- Implementation of:
- Inorder Traversal
- Preorder Traversal
- Postorder Traversal
- Level Order Traversal
- Binary Search Tree
- AVL Tree
- Red Black Tree
- Practice Problems

Graphs

- Introduction to Graph
- Graph Representation
- Breadth-First Search
- Depth First Search
- Prim's Algorithm/Minimum Spanning Tree
- Dijkstra's Shortest Path Algorithm
- Bellman-Ford Shortest Path Algorithm
- Kruskal's Algorithm
- Floyd Warshall Algorithm
- Practice Problems

Greedy

- Introduction
- Activity Selection Problem
- Fractional Knapsack
- Job Sequencing Problem
- Huffman Coding
- Practice Problems

Backtracking

- Concepts of Backtracking
- N Queen Problem
- Sudoku Problem
- Practice Problems

Dynamic Programming

- Introduction
- Dynamic Programming
- Memoization
- Tabulation
- Practice Problems

Trie

- Introduction
- Representation
- Search
- Insert
- Delete
- Count Distinct Rows in a Binary Matrix
- Practice Problems



Reach us at:

info@drixcat.com | +91-9953356262

Placement Partners:

